FPGA Implementation

Experiment : 07

EE2801: DSP Lab

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Contents

1 Experiment Objectives

Implementing

- FIR filter, non-pipelined architecture
- FIR filter using a pipelined architecture
- 8 bit FFT using a radix-2

All modules use fixed-point arithmetic in Q(2,14) format, giving 17 total bits. This same width is used across the FIR coefficients, ROM data, board-level sample bus, and Signal Tap observations so that the flow remains consistent for all three experiments.

2 FPGA Implementation and Board Bring-Up Procedure

This section describes the complete procedure to configure the FPGA board, generate input data, compile the design in Quartus Prime, and verify outputs using SignalTap.

2.1 Project Setup

- Create a project directory named `fir_single_mac`.
- Initialize a Quartus Prime project with the same name.
- Select FPGA device: **Cyclone V (5CSEBA6U23I7)**.
- Organize the project directory as follows:
  - `src/rtl` - RTL design files
  - `src/tb` - Testbench files
  - `data` - Input/output data files
  - `ip` - Generated IP cores
  - `scripts` - Python scripts

2.2 Fixed-Point Representation

All signals are represented in **Q(2,14)** format (17-bit signed fixed-point representation).

2.3 Coefficient and Data Generation

- A Python script (`coeff_gen.py`) is used to generate FIR filter coefficients as a Verilog module.
- Input signal data is generated using a Python script and stored as `.mif` files.
- The memory initialization file follows:

```
WIDTH=17;
DEPTH=1000;
ADDRESS_RADIX=UNS;
DATA_RADIX=HEX;
CONTENT BEGIN
    0: data0
    ...
END;
```


- Signals are generated for frequencies: 900 Hz, 1100 Hz, and 2000 Hz.
- Sampling frequency is fixed at 100 kHz for correct operation.

2.4 ROM IP Creation

- Create a **1-Port ROM** using Quartus IP Catalog.
- Configuration:
  - Data width: 17 bits
  - Depth: 10000
  - Single clock operation
- Load generated `.mif` file as memory initialization.
- Enable *In-System Memory Content Editor*.

2.5 RTL Integration

- Implement the following modules:
  - FIR Filter (Non-pipelined / Pipelined)
  - Data Generator (ROM wrapper)
  - Top-level wrapper (`de10_nano_top`)
- The data generator feeds samples to the FIR filter based on ready/busy signals.
- Use `(* keep *)` attributes to prevent signal optimization.

2.6 Timing Constraints

- Create an SDC file specifying:

```
create_clock -name clk -period 20 [get_ports FPGA_CLK]
```

- This sets the clock frequency to 50 MHz.

2.7 Pin Configuration

Using Pin Planner:

- `KEY[0]` (Reset) → PIN_AH17
- `FPGA_CLK` → PIN_V11
- I/O standard: **3.3V LVTTL**

2.8 Programming the FPGA

- Connect DE10-Nano board via USB Blaster.
- Open Quartus Programmer.
- Click `Auto Detect` and select device `5CSEBA6`.
- Load the generated `.sof` file.
- Start programming.
- Verify that the `CONF_DONE` LED is ON.

2.9 SignalTap Analysis

- Open SignalTap Logic Analyzer.
- Select clock signal from design.
- Add nodes:
  - Input signal (from data generator)

  - Output signal (from FIR filter)
  - Ready signal
- Group signals as buses (MSB to LSB order).
- Set:
  - Sample depth: 8K
  - Trigger: Rising edge of ready signal
- Compile and reprogram FPGA.
- Run SignalTap to capture waveform.

2.10 Result Extraction

- Convert captured signals to signed decimal format.
- Export waveform data as `.csv`.
- Use Python scripts to plot input and output signals.

2.11 Testing Procedure

- Load different input signals (900 Hz, 1100 Hz, 2000 Hz).
- Observe filtering behavior for:
  - Non-pipelined FIR
  - Pipelined FIR
  - Feedforward structure
- Capture and compare output responses.

3 FIR Non Pipelined

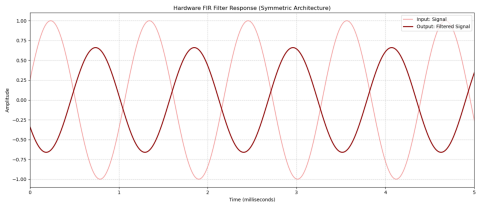

Fig. 0.1: 900Hz

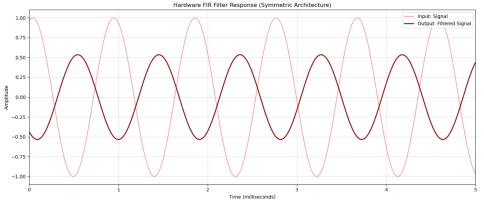

Fig. 0.2: 1100Hz

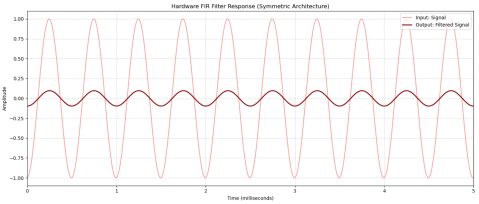

Fig. 0.3: 2000Hz

4 FIR Pipelined

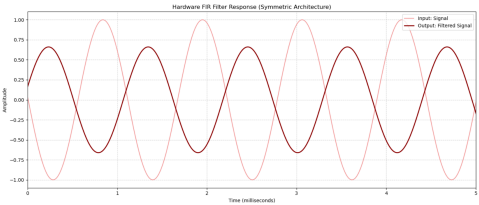

Fig. 0.4: 900Hz

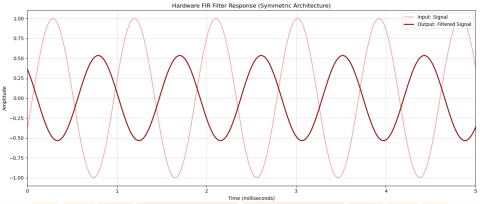

Fig. 0.5: 1100Hz

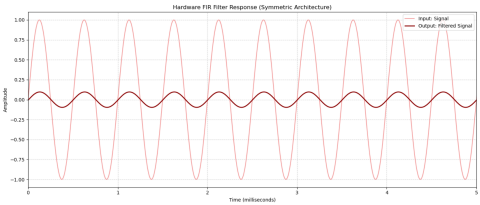

Fig. 0.6: 2000Hz

5 FFT Pipelined

Flow Summary	
<<Filter>>	
Flow Status	Successful - Sun Apr 12 15:29:18 2026
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	fftharshil
Top-level Entity Name	fft_top
Family	Arria II GX
Logic utilization	3 %
Total registers	642
Total pins	42 / 176 (24 %)
Total virtual pins	0
Total block memory bits	0 / 2,939,904 (0 %)
DSP block 18-bit elements	0 / 232 (0 %)
Total GXB Receiver Channel PCS	0 / 4 (0 %)
Total GXB Receiver Channel PMA	0 / 4 (0 %)
Total GXB Transmitter Channel PCS	0 / 4 (0 %)
Total GXB Transmitter Channel PMA	0 / 4 (0 %)
Total PLLs	0 / 4 (0 %)
Total DLLs	0 / 2 (0 %)
Device	EP2AGX45CU17I3
Timing Models	Final